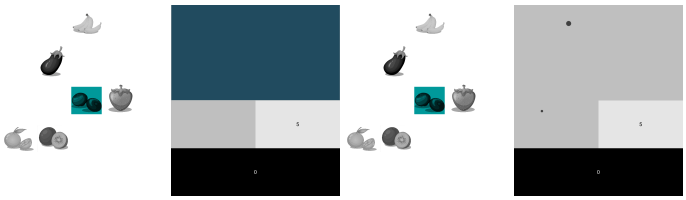
Iconicity has benefits for bootstrapping, but they're not only for the audience

Gareth Roberts*[1], Robert Snider[1,2], and Petros Kaklamanis[1]

*Corresponding Author: gareth.roberts@ling.upenn.edu
[1]Department of Linguistics, University of Pennsylvania, Philadelphia, USA
[2]Department of Theoretical and Applied Linguistics, University of Cambridge, Cambridge, UK

Iconicity has long been observed to be widespread in newly emerging sign languages (e.g., Sandler et al., 2011) and in experiments on the emergence of communication systems (e.g., Verhoef et al., 2016; Roberts et al., 2015). An obvious inference is that iconicity aids in establishing reference. The typical mechanism assumed is *semantic transparency*: iconic signals make it easier for the audience to identify the intended meaning (Lockwood et al., 2016; Auch et al., 2020). But this audience-based account ignores potential *production-side* benefits. Iconic forms may come more readily to mind, for instance, or be recalled more easily, providing a stable communicative anchor in repeated interactions.



(a) *High or One-sided iconicity condition* (b) *Low iconicity condition.*

Figure 1. Example screenshots of Sender's screen in different conditions.

To our knowledge, no work has fully decoupled production- from audience-side iconicity in a communicative setting. We investigated this using a communication game paradigm, based on Roberts and Clark (2020). In our game a Sender moved their finger on a trackpad to select signals to communicate six different fruit and vegetable referents to a Receiver. Sender and Receiver roles remained constant throughout, and we manipulated the level of iconicity in signals (Fig. 1). In the *High iconicity* condition, the Sender selected colors from an underlying colorspace, with different colors corresponding to different finger positions; the Receiver saw these colors. In the *Low iconicity* condition, neither participant saw colors; the Sender simply sent their finger location on the trackpad (represented as

a dot in a rectangle) to the Receiver. Finally, in the *One-sided iconicity* condition, the Sender sent a color, but the Receiver instead saw the Sender's finger position as a dot; neither participant was aware of the difference in experience.

We operationalized establishing a signal for a referent as communicating the referent successfully for $>= 75\%$ of its last four occurrences. High-iconicity dyads established signals for all referents quickly ($M = 10.05$ min); dyads in the Low iconicity condition took the longest ($M = 32.8$ min.) and had more difficulty establishing signals than in either of the other conditions. The One-sided iconicity condition is the most important, however. Fig. 2 shows the proportion of rounds dyads spent with different numbers of established signals. This suggests One-sided iconicity dyads experienced a head-start relative to Low iconicity dyads. Binomial mixed effects models showed significant main effects of condition (for One-sided and High iconicity) and round number, and an interaction between round number and condition, both for establishing signals, $\beta = -0.496, SE = 0.09, z = -5.26, p < 0.001$ and for number of signals, $\beta = 2.03, SE = 0.22, z = 9.08, p < 0.001$.[1]

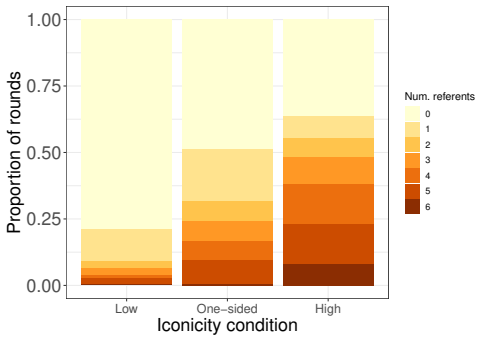


Figure 2. Proportion of rounds with each signal set size across all dyads, organized by condition (practice rounds included).

This head-start was driven by greater Sender-side stability. We measured the distance between successive signals for the same referent; lower distances imply greater stability. These did not differ in the High- and One-sided iconicity conditions ($p = 0.37$), but were significantly higher for Low iconicity $\beta = -0.07, SE = 0.02, t(65.85) = -3.1, p = 0.0029$ (compared with One-sided).

Our results suggest the bootstrapping benefits of iconicity are not purely audience-based, but also involve distinct benefits for the signal producer.

[1]First model coded referent absence and presence as 0 and 1; second was performed on referent numbers above 0 only. Both had condition (Low iconicity as reference level) and round number as predictors, with random intercepts for dyads.

References

Auch, L., Gagné, C. L., & Spalding, T. L. (2020). Conceptualizing semantic transparency: A systematic analysis of semantic transparency measures in English compound words. *Methods in Psychology*, *3*, 100030.

Lockwood, G., Hagoort, P., & Dingemanse, M. (2016). How iconicity helps people learn new words: Neural correlates and individual differences in sound-symbolic bootstrapping. *Collabra*, *2*(1), 7.

Roberts, G., & Clark, R. (2020). Dispersion, communication, and alignment: An experimental study of the emergence of structure in combinatorial phonology. *Journal of Language Evolution*, *5*(2), 121–139.

Roberts, G., Lewandowski, J., & Galantucci, B. (2015). How communication changes when we cannot mime the world: Experimental evidence for the effect of iconicity on combinatoriality. *Cognition*, *141*, 52–66.

Sandler, W., Aronoff, M., Meir, I., & Padden, C. (2011). The gradual emergence of phonological form in a new language. *Natural Language & Linguistic Theory*, *29*, 503–543.

Verhoef, T., Kirby, S., & De Boer, B. (2016). Iconicity and the emergence of combinatorial structure in language. *Cognitive Science*, *40*(8), 1969–1994.